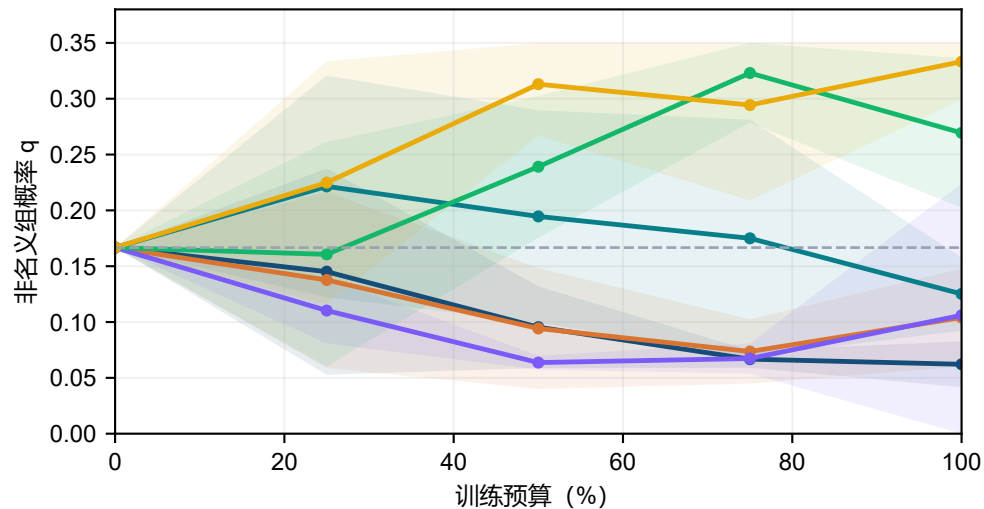
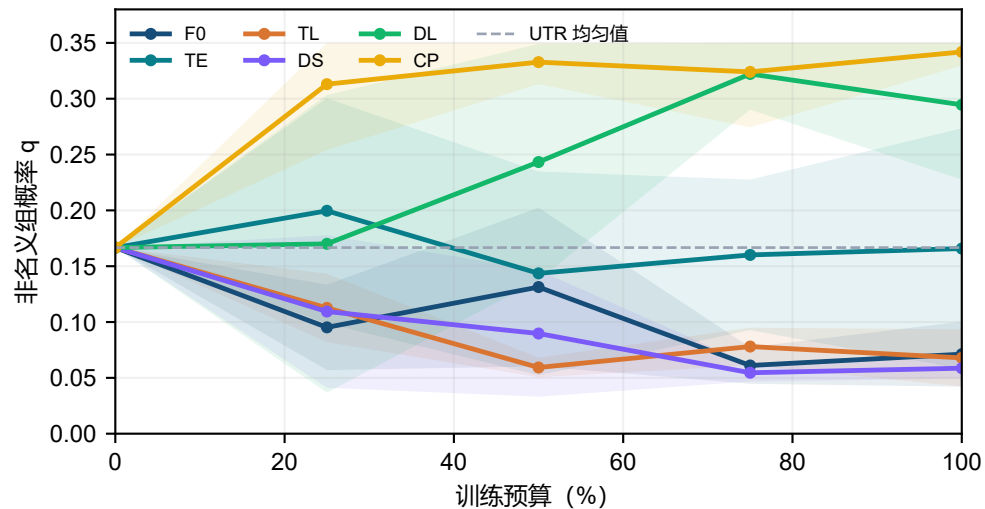


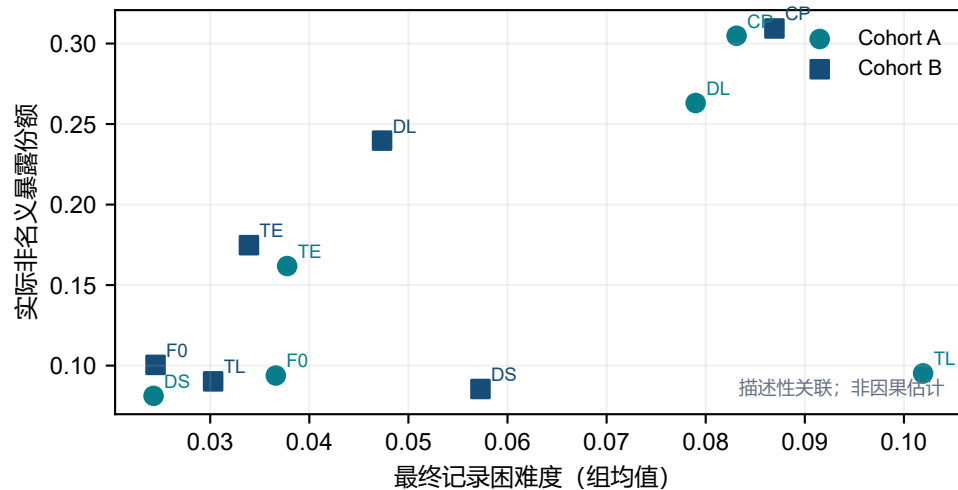
Cohort A: 冻结 sampler q 轨迹 (n=5)



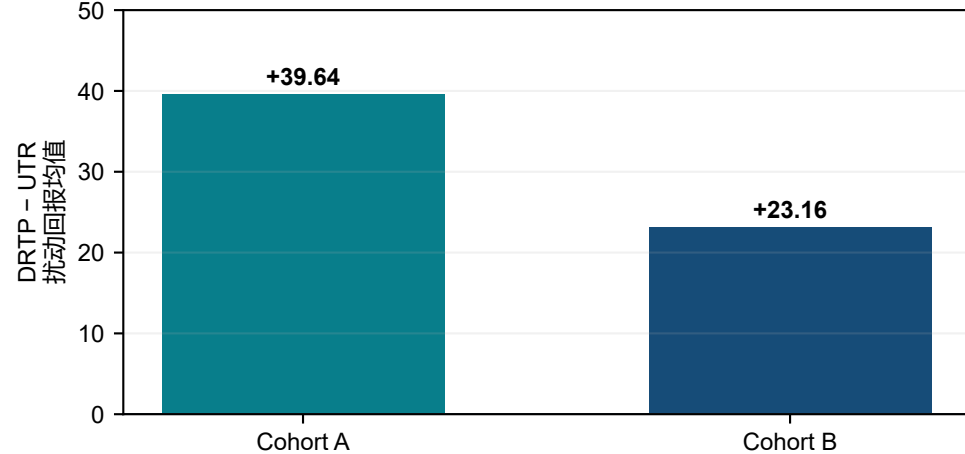
Cohort B: 冻结 sampler q 轨迹 (n=5)



组级困难度与实际训练暴露



冻结终点协同结果



真实 q 轨迹、实际暴露与困难度均来自 SHA256 验证的最终训练包; manifest 未归档显式邻接/边列表, 故本图不将故障组标签替代为连通性或最短路径统计。